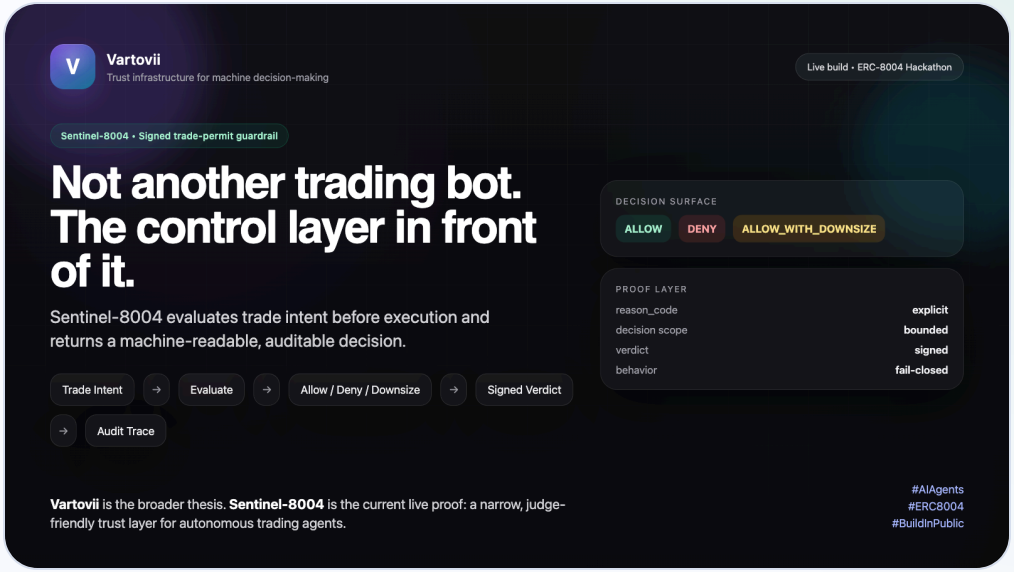
PROBLEM

# Autonomous trading agents need a control layer before capital moves.

Agents can act faster than humans, but speed without bounded execution scope creates failure modes that are hard to audit after the fact.

> **Sentinel-8004 focuses on the last safe checkpoint before execution.**




### Before execution

Evaluate the trade intent before the venue or chain call happens.

### Fail closed

Block when required inputs or safe execution proof are missing.

### Machine readable

Return structured outputs judges can inspect without private systems.

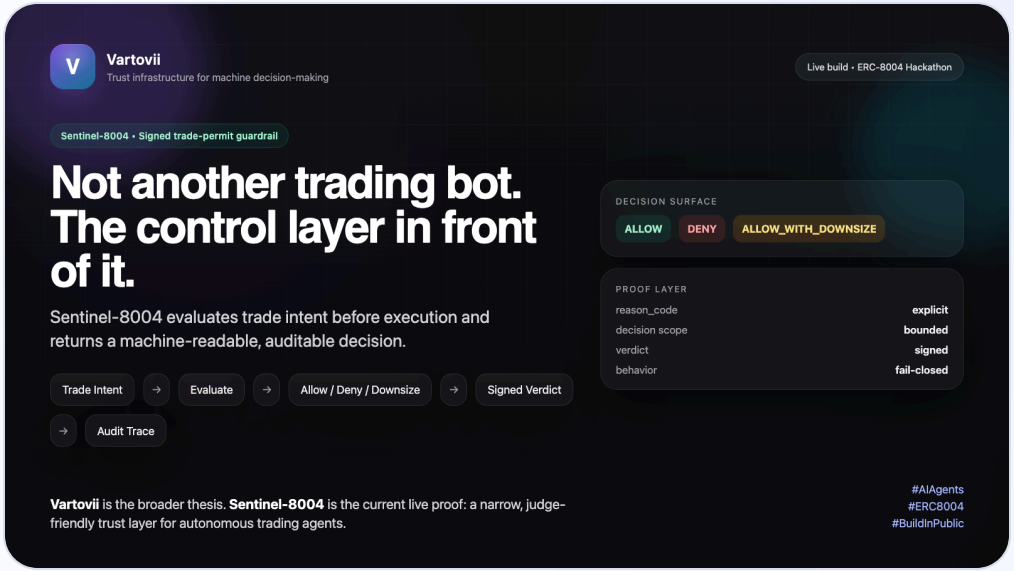
THESIS

# Sentinel-8004 is not another trading bot.

- **Guardrail first:** approve, deny, or downsize a proposed trade.
- **Bounded execution:** attach a signed permit envelope to the verdict.
- **Auditable proof:** surface a validation artifact that binds registration, decision hash, and permit hash.

**POSITIONING**

We do not build bots for capital deployment. We build guardrails for bots that move capital.



PRODUCT FLOW

# One narrow path from intent to execution gate.

The public repo stays intentionally narrow: evaluate the trade, emit the proof objects, and verify that the requested execution remains within the approved envelope.

### 1. Trade Intent

Agent, venue, market, side, size, and required signals.

### 2. Policy Evaluation

Configured checks return ALLOW, DENY, or ALLOW_WITH_DOWNSIZE.

### 3. Signed Verdict

Decision hash, permit hash, signer, expiry, and scope payload.

### 4. Validation Artifact

Registration binding plus proof status: VALIDATED, CONSTRAINED, or BLOCKED.

### 5. Permit Verification

Execution is allowed only if the request still fits the signed envelope.

PUBLIC-SAFE BOUNDARY

- No exchange execution, private bridge logic, or hidden operator surface.
- Judges can inspect the exact JSON objects that drive the guardrail story.
- When the repo cannot prove a safe path, the demo stays fail closed.

PROOF ARTIFACTS

# Three objects make the control story inspectable.

SIGNED VERDICT

## Execution decision

Machine-readable output that records the verdict and the bounded permit envelope.

decision_hash · permit_hash · signature · expires_at

VALIDATION ARTIFACT

## ERC-8004-facing proof layer

Public-safe artifact that binds registration scope, verdict scope, and permit scope for the demo.

registration_id · proof_status · artifact_hash · demo_only

PERMIT VERIFICATION

## Last execution gate

Deterministic check that the requested execution still stays inside the approved notional and scope.

permit_valid · executable · verification_code · checks[]

DEMO-ONLY BOUNDARY

The proof artifacts are real public demo objects, but they do not claim live on-chain registry verification or live capital execution.

LIVE DEMO SURFACE

# Hosted root hub plus one canonical judge walkthrough.

The hosted app entrypoint stays simple. Judges land on the hub, then move into the live demo at **/judge**.

JUDGE MODE DEMO SHELL

## Sentinel-8004 public proof walkthrough

Pick one canonical scenario and inspect the exact public-safe payloads that drive the guardrail story.

Scenario

allow-btc-buy | Reload

Scenario: allow-btc-buy | Verdict: ALLOW | Proof: VALIDATED | Permit executable: yes

### Trade Intent


```
{
  "trace_id": "trace-allow-btc-buy-001",
  "agent_id": "strategy-agent-demo",
  "intent_id": "intent-allow-btc-buy-001",
  "venue": "kraken",
  "chain_id": 1,
  "market": "BTC/USD",
  "side": "BUY",
  "order_type": "MARKET",
  "size_base": "0.0500",
  "notional_usd": "2400.00",
  "max_slippage_bps": 35,
  "strategy_context": {
    "demo_scenario": "allow",
    "thesis": "Momentum confirmation with approved size and asset allowlist match.",
    "required_signals": {
      "market_data": "available",
      "position_limit": "available"
    }
  },
  "submitted_at": "2026-03-27T09:00:00Z"
}
```


### Verdict


```
{
  "trace_id": "trace-allow-btc-buy-001",
  "verdict": "ALLOW",
  "risk_score": 20,
  "reason_code": "RISK_WITHIN_POLICY",
  "reason_detail": [
    "market_allowlisted",
    "requested_notional_within_policy_cap",
    "required_signals_available"
  ],
  "allowed_notional_usd": "2400.00",
  "expires_at": "2026-03-27T09:05:00.000Z",
  "policy_version": "judge-demo-v1",
  "judge_mode": true,
  "decision_hash": "0xf41f785c1363a4e3c2022173629c2681675468a38ed5d55d61281df4f97df630",
  "signature": "0xa077caef622f01247184050 65c37afb8bb013a2e39e6743b9e096b1b8bd5abed",
  "signer": "sentinel-demo-signer",
  "signed_at": "2026-03-27T09:00:00Z",
  "expires_at": "2026-03-27T09:05:00.000Z",
  "schema_version": "sentinel-8004-v1",
  "verdict_payload": {
```


### Validation Artifact


```
{
  "artifact_id": "validation-intent-allow-btc-buy-001",
  "trace_id": "trace-allow-btc-buy-001",
  "intent_id": "intent-allow-btc-buy-001",
  "agent_id": "strategy-agent-demo",
  "registration_id": "agent-reg-strategy-agent-demo-001",
  "registration_status": "ACTIVE",
  "registration_hash": "0x4b05fa67bf7ba32100130 5eb5d2729d95facae0dbdfc00b938da50435e43d8fa",
  "verdict": "ALLOW",
  "allowed_notional_usd": "2400.00",
  "decision_hash": "0xf41f785c1363a4e3c2022173629c2681675468a38ed5d55d61281df4f97df630",
  "permit_hash": "0x706ba65764be631b02377fe95c2c25e3a1cdd20412f2c5e1bc07a0df6c2ee070",
  "proof_status": "VALIDATED",
  "proof_checks": [
    "agent_registration_active",
    "chain_in_registration_scope",
    "venue_in_registration_scope",
    "market_in_registration_scope",
    "decision_hash_bound",
    "permit_hash_bound",
    "execution_permitted"
  ],
  "created_at": "2026-03-27T09:00:00Z",
  "expires_at": "2026-03-27T09:05:00.000Z",
  "schema_version": "sentinel-8004-validation-artifact-v1",
  "demo_only": true,
  "artifact_hash": "0x0ac3a539da4d5a1146488e6534c1b664a1bee3c82e59c3846416f3eb3395fddd"
}
```


### Permit Verification


```
{
  "trace_id": "trace-allow-btc-buy-001",
  "permit_valid": true,
  "executable": true,
  "verification_code": "EXECUTION_PERMITTED",
  "requested_notional_usd": "2400.00",
  "approved_notional_usd": "2400.00",
  "decision_hash": "0xf41f785c1363a4e3c2022173629c2681675468a38ed5d55d61281df4f97df630",
  "permit_hash": "0x706ba65764be631b02377fe95c2c25e3a1cdd20412f2c5e1bc07a0df6c2ee070",
  "verified_at": "2026-03-27T09:00:00Z",
  "judge_mode": true,
  "checks": [
    {
      "name": "trace_id_matches",
      "ok": true,
      "detail": "Trace identifiers stay aligned across the intent and signed permit envelope."
    },
    {
      "name": "decision_hash_matches",
      "ok": true,
      "detail": "Signed decision hash matches the canonical verdict payload."
    },
    {
      "name": "permit_scope_matches",
      "ok": true,
      "detail": "Permit payload matches the submitted execution intent envelope."
    },
    {
```


**Approved path:** ALLOW verdict, VALIDATED proof status, and executable permit.

JUDGE MODE DEMO SHELL

## Sentinel-8004 public proof walkthrough

Pick one canonical scenario and inspect the exact public-safe payloads that drive the guardrail story.

Scenario

downsize-eth-buy | Reload

Scenario: downsize-eth-buy | Verdict: ALLOW_WITH_DOWNSIZE | Proof: CONSTRAINED | Permit executable: no

### Trade Intent


```
{
  "trace_id": "trace-downsize-eth-buy-001",
  "agent_id": "strategy-agent-demo",
  "intent_id": "intent-downsize-eth-buy-001",
  "venue": "kraken",
  "chain_id": 1,
  "market": "ETH/USD",
  "side": "BUY",
  "order_type": "MARKET",
  "size_base": "1.2000",
  "notional_usd": "3200.00",
  "max_slippage_bps": 40,
  "strategy_context": {
    "demo_scenario": "downsize",
    "thesis": "Direction is acceptable, but requested exposure is above the preferred demo threshold.",
    "required_signals": {
      "market_data": "available",
      "position_limit": "available"
    }
  },
  "submitted_at": "2026-03-27T09:10:00Z"
}
```


### Verdict


```
{
  "trace_id": "trace-downsize-eth-buy-001",
  "verdict": "ALLOW_WITH_DOWNSIZE",
  "risk_score": 61,
  "reason_code": "POSITION_SIZE_EXCEEDS_LIMIT",
  "reason_detail": [
    "requested_notional_above_soft_cap",
    "downsized_to_policy_cap"
  ],
  "allowed_notional_usd": "2500.00",
  "expires_at": "2026-03-27T09:15:00.000Z",
  "policy_version": "judge-demo-v1",
  "judge_mode": true,
  "decision_hash": "0xfe4c2b9eab1f2c7e9280f1f7900a211c12e2528016152be3fef913b9edc9110",
  "signed_verdict": {
    "trace_id": "trace-downsize-eth-buy-001",
    "decision_hash": "0xfe4c2b9eab1f2c7e9280f1f7900a211c12e2528016152be3fef913b9edc9110",
    "permit_hash": "0xad2b2ab110a442e1bda502fbece0836d78c77f91ee5b43a9896622f6a13a41f7",
    "signature": "0x1a4e9aa6329b9c729bbf1bbcb30473f5ae00e8a900225d625db0509f0df94b49",
    "signer": "sentinel-demo-signer",
    "signed_at": "2026-03-27T09:10:00Z",
    "expires_at": "2026-03-27T09:15:00.000Z",
    "schema_version": "sentinel-8004-v1",
    "verdict_payload": {
      "trace_id": "trace-downsize-
```


### Validation Artifact


```
{
  "artifact_id": "validation-intent-downsize-eth-buy-001",
  "trace_id": "trace-downsize-eth-buy-001",
  "intent_id": "intent-downsize-eth-buy-001",
  "agent_id": "strategy-agent-demo",
  "registration_id": "agent-reg-strategy-agent-demo-001",
  "registration_status": "ACTIVE",
  "registration_hash": "0x4b05fa67bf7ba32100130 5eb5d2729d95facae0dbdfc00b938da50435e43d8fa",
  "verdict": "ALLOW_WITH_DOWNSIZE",
  "allowed_notional_usd": "2500.00",
  "decision_hash": "0xfe4c2b9eab1f2c7e9280f1f7900a211c12e2528016152be3fef913b9edc9110",
  "permit_hash": "0xad2b2ab110a442e1bda502fbece0836d78c77f91ee5b43a9896622f6a13a41f7",
  "proof_status": "CONSTRAINED",
  "proof_checks": [
    "agent_registration_active",
    "chain_in_registration_scope",
    "venue_in_registration_scope",
    "market_in_registration_scope",
    "decision_hash_bound",
    "permit_hash_bound",
    "downsized_execution_only"
  ],
  "created_at": "2026-03-27T09:10:00Z",
  "expires_at": "2026-03-27T09:15:00.000Z",
  "schema_version": "sentinel-8004-validation-artifact-v1",
  "demo_only": true,
  "artifact_hash": "0xeaf8f7dc18853230d670e3858c1bf3c0186bdae1d2df4f49731ede7bfb9c55f5"
}
```


### Permit Verification


```
{
  "trace_id": "trace-downsize-eth-buy-001",
  "permit_valid": true,
  "executable": false,
  "verification_code": "REQUEST_EXCEEDS_APPROVED_NOTIONAL",
  "requested_notional_usd": "3200.00",
  "approved_notional_usd": "2500.00",
  "decision_hash": "0xfe4c2b9eab1f2c7e9280f1f7900a211c12e2528016152be3fef913b9edc9110",
  "permit_hash": "0xad2b2ab110a442e1bda502fbece0836d78c77f91ee5b43a9896622f6a13a41f7",
  "verified_at": "2026-03-27T09:10:00Z",
  "judge_mode": true,
  "checks": [
    {
      "name": "trace_id_matches",
      "ok": true,
      "detail": "Trace identifiers stay aligned across the intent and signed permit envelope."
    },
    {
      "name": "decision_hash_matches",
      "ok": true,
      "detail": "Signed decision hash matches the canonical verdict payload."
    },
    {
      "name": "permit_scope_matches",
      "ok": true,
      "detail": "Permit payload matches the submitted execution intent envelope."
    },
```


**Constrained path:** ALLOW_WITH_DOWNSIZE verdict, CONSTRAINED proof status, and a blocked oversized execution request.

WHY THIS MATTERS

# Autonomous trading agents need bounded decisions, not blind execution.

- **Clear control point:** a single guardrail layer before capital movement.
- **Inspection friendly:** judges can trace intent, verdict, validation artifact, and permit verification in one public repo.
- **Practical next step:** the same pattern can front-run live agent execution paths without turning this repo into a generic platform.

**GitHub repo:** github.com/Vetassikc/Vartovii-Sentinel-8004

**Hosted entrypoint:** root URL -> submission hub

**Live walkthrough:** /judge

JUDGE TAKEAWAY

Sentinel-8004 is a signed trade-permit guardrail for autonomous trading agents.